# The truncation of a path type is the path type of the truncation, for every truncation level

The Univalent Foundations Program

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*Editorial note.* The counted statement is the author's thm environment at source lines 671-674 and its proof environment opens at line 676 after a blank line; the pairing is by adjacency and the retained proof range 676-722 is the whole environment.

*Editorial note.* The retained counter values (chapter 7, section 3, theorem offset 11) make the counted theorem print as Theorem 7.3.12, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel thm:path-truncation = 7.3.12).

*Editorial note.* Four labels the statement and proof cite lie outside the card and are declared in external_references as non-load-bearing navigation pointers: eq:path-trunc-map (the equation defining the map whose equivalence is counted, book equation 7.3.11, defined at line 664 of the same pinned file), thm:hleveln-of-hlevelSn (book Theorem 7.1.11, defined at line 220 of the same file), thm:transport-compose (2.3.10) and thm:ap-transport (2.3.11); the last two are results of chapter 2, whose file is not part of the pinned corpus, so their pointer spans are the label lines of this chapter's label-number dump.

*Editorial note.* The retained preamble is the book's own macro library and layout adapter as stored for this chapter; no package outside it is requested.

*Editorial scope.* One original result of the pinned HoTT Book source tree, counted once: the equivalence between the truncation of a path type and the path type of the truncation (author label thm:path-truncation, book Theorem 7.3.12).

*Allowed standard prerequisites (editorial).*

- Truncation and its induction principle. \trunc n A is the n-truncation of A with \tproj n{\blank} : A \to \trunc n A, and it is universal among n-truncated types.. It is the operation whose interaction with path types is computed.
- Encode-decode method. To show a map into a path type of a truncated type is an equivalence, generalise the two endpoints and build a family P on the truncation together with maps decode and encode that are inverse to each other.. It is the method the whole proof follows (source lines 677-679).

- Level of a type of truncated types. The type \typele{n} of n-truncated types is itself (n+1)-truncated, so a family with values in it can be defined by induction from its value at the base point.. It is the justification that P is well defined (source line 686).

*Source reference.* Original equation 7.3.11: sources/hott-hlevels/pinned-original-source.tex, source lines 664–664. The displayed equation that defines the map whose being an equivalence is the counted theorem (the map f : \trunc n{x=_A y} \to (\tproj{n+1}x = \tproj{n+1}y)). It is a navigation pointer: the retained proof reconstructs the same map through decode and encode, and nothing of the argument is delegated to the pointer span itself.

*Source reference.* Original theorem 7.1.11: sources/hott-hlevels/pinned-original-source.tex, source lines 220–220. Book Theorem 7.1.11, the level of the type of n-truncated types, cited at source line 686 to justify that the family P is well defined. Navigation pointer only: the reason it supplies is stated in the retained text itself and the counted proof is retained in full.

*Source reference.* Original theorem 2.3.10: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1894–1894. Book Theorem 2.3.10, the composition rule for transport, used in the last computation of the retained proof. Navigation pointer only; its defining chapter is not part of the pinned corpus, so the pointer span is the label line of this chapter's label-number dump.

*Source reference.* Original theorem 2.3.11: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1750–1750. Book Theorem 2.3.11, the commutation of transport with the action on paths, used in the same computation. Navigation pointer only; its defining chapter is not part of the pinned corpus, so the pointer span is the label line of this chapter's label-number dump.

**Theorem 7.3.12.** *For any $A$ and $x, y : A$ and $n \geq -2$, the map* (7.3.11) *is an equivalence; thus we have*

$$\left\|x =_A y\right\|_n \simeq \left(|x|_{n+1} =_{\|A\|_{n+1}} |y|_{n+1}\right).$$

*Proof.* The proof is a simple application of the encode-decode method: As in previous situations, we cannot directly define a quasi-inverse to the map (7.3.11) because there is no way to induct on an equality between $|x|_{n+1}$ and $|y|_{n+1}$. Thus, instead we generalize its type, in order to have general elements of the type $\|A\|_{n+1}$ instead of $|x|_{n+1}$ and $|y|_{n+1}$. Define $P : \|A\|_{n+1} \to \|A\|_{n+1} \to n\text{-}\mathsf{Type}$ by

$$P(|x|_{n+1}, |y|_{n+1}) :\equiv \|x =_A y\|_n$$

This definition is correct because $\|x =_A y\|_n$ is $n$-truncated, and $n$-$\mathsf{Type}$ is $(n+1)$-truncated by original theorem 7.1.11. Now for every $u, v : \|A\|_{n+1}$, there is a map

$$\mathsf{decode} : P(u, v) \to \left(u =_{\|A\|_{n+1}} v\right)$$

defined for $u = |x|_{n+1}$ and $v = |y|_{n+1}$ and $p : x = y$ by

$$\mathsf{decode}(|p|_n) :\equiv \mathsf{ap}_{|-|_{n+1}}(p).$$

Since the codomain of $\mathsf{decode}$ is $n$-truncated, it suffices to define it only for $u$ and $v$ of this form, and then it's just the same definition as before. We also define a function

$$r : \prod_{u:\|A\|_{n+1}} P(u,u)$$

by induction on $u$, where $r(|x|_{n+1}) :\equiv |\mathsf{refl}_x|_n$.

Now we can define an inverse map

$$\mathsf{encode} : (u =_{\|A\|_{n+1}} v) \to P(u,v)$$

by

$$\mathsf{encode}(p) :\equiv \mathsf{transport}^{v\mapsto P(u,v)}(p, r(u)).$$

To show that the composite

$$(u =_{\|A\|_{n+1}} v) \xrightarrow{\mathsf{encode}} P(u,v) \xrightarrow{\mathsf{decode}} (u =_{\|A\|_{n+1}} v)$$

is the identity function, by path induction it suffices to check it for $\mathsf{refl}_u : u = u$, in which case what we need to know is that $\mathsf{decode}(r(u)) = \mathsf{refl}_u$. But since this is an $(n-1)$-type, hence also an $(n+1)$-type, we may assume $u \equiv |x|_{n+1}$, in which case it follows by definition of $r$ and $\mathsf{decode}$. Finally, to show that

$$P(u,v) \xrightarrow{\mathsf{decode}} (u =_{\|A\|_{n+1}} v) \xrightarrow{\mathsf{encode}} P(u,v)$$

is the identity function, since this goal is again an $(n-1)$-type, we may assume that $u = |x|_{n+1}$ and $v = |y|_{n+1}$ and that we are considering $|p|_n : P(|x|_{n+1}, |y|_{n+1})$ for some $p : x = y$. Then we have

$$\begin{aligned}
\mathsf{encode}(\mathsf{decode}(|p|_n)) &= \mathsf{encode}(\mathsf{ap}_{|-|_{n+1}}(p)) \\
&= \mathsf{transport}^{v\mapsto P(|x|_{n+1},v)}(\mathsf{ap}_{|-|_{n+1}}(p), |\mathsf{refl}_x|_n) \\
&= \mathsf{transport}^{y\mapsto \|x=y\|_n}(p, |\mathsf{refl}_x|_n) \\
&= \left|\mathsf{transport}^{y\mapsto (x=y)}(p, \mathsf{refl}_x)\right|_n \\
&= |p|_n,
\end{aligned}$$

using original theorems 2.3.10 and 2.3.11. (Alternatively, we could do path induction on $p$; the desired equality would then hold judgmentally.) This completes the proof that $\mathsf{decode}$ and $\mathsf{encode}$ are quasi-inverses. The stated result is then the special case where $u = |x|_{n+1}$ and $v = |y|_{n+1}$. □